

Estadística Actuarial Practica 3 Segunda Parte

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En ATENEA descarga la base de datos dataclaim.csv, que contiene los datos siguientes:

Variables	Descripcion
veh_value	valor del vehículo, en \$10,000s
exposure	fracción del año con cobertura
numclaims	número de siniestros durante el período de cobertura
claimcst0	coste total de los siniestros (0 si no los hay)
veh_body	tipo de vehículo
veh_age	1 (vehículos más jóvenes), 2, 3, 4 (vehículos más antiguos)
gender	F = femenino, M = masculino
area	área de circulación: A, B, C, D, E y F
agecat	1 (conductor más joven), 2, 3, 4, 5, 6 (conductor más mayor)

Realiza los siguientes cálculos:

- Haz un descriptivo de los datos.
- Ajusta un modelo de regresión de Poisson y otro modelo Binomial Negativo para la frecuencia de siniestralidad. Interpreta los resultados. ¿Cuál te parece más adecuado?
- Ajusta un modelo de regresión Gamma, Inversa Gausiana y Lognormal a la severidad de los siniestros. Interpreta los resultados. ¿Cuál modelo te parece mejor?
- Calcula para cada asegurado la prima pura para el año que viene.

Lectura y preparación de los datos:

```
datos <- read.csv("dataclaim.csv", stringsAsFactors = FALSE)
```

```
str(datos)
```

```
## 'data.frame':    67856 obs. of  9 variables:
## $ veh_value: num  1.06 1.03 3.26 4.14 0.72 2.01 1.6 1.47 0.52 0.38 ...
## $ exposure : num  0.304 0.649 0.569 0.318 0.649 ...
## $ numclaims: int   0 0 0 0 0 0 0 0 0 0 ...
## $ claimcst0: num   0 0 0 0 0 0 0 0 0 0 ...
## $ veh_body : chr  "HBACK" "HBACK" "UTE" "STNWG" ...
## $ veh_age  : int   3 2 2 2 4 3 3 2 4 4 ...
## $ gender   : chr  "F" "F" "F" "F" ...
## $ area     : chr  "C" "A" "E" "D" ...
## $ agecat   : int   2 4 2 2 2 4 4 6 3 4 ...
```

```
summary(datos)
```

```
##    veh_value      exposure      numclaims      claimcst0
## Min.   : 0.000    Min.   :0.002738    Min.   :0.00000    Min.   :  0.0
## 1st Qu.: 1.010    1st Qu.:0.219028    1st Qu.:0.00000    1st Qu.:  0.0
```

```
## Median : 1.500    Median :0.446270    Median :0.00000    Median : 0.0
## Mean : 1.777     Mean :0.468652    Mean :0.07276    Mean : 137.3
## 3rd Qu.: 2.150    3rd Qu.:0.709103    3rd Qu.:0.00000    3rd Qu.: 0.0
## Max. :34.560     Max. :0.999316    Max. :4.00000    Max. :55922.1
## veh_body          veh_age          gender          area
## Length:67856      Min. :1.000    Length:67856      Length:67856
## Class :character   1st Qu.:2.000    Class :character   Class :character
## Mode :character    Median :3.000    Mode :character    Mode :character
##                    Mean :2.674
##                    3rd Qu.:4.000
##                    Max. :4.000
## agecat
## Min. :1.000
## 1st Qu.:2.000
## Median :3.000
## Mean :3.485
## 3rd Qu.:5.000
## Max. :6.000
```

```
# Variables categóricas
vars_factor <- c("veh_body", "veh_age", "gender", "area", "agecat")

datos[vars_factor] <- lapply(datos[vars_factor], factor)

# Eliminamos exposiciones nulas si existen

datos <- subset(datos, exposure > 0)

# Variable de severidad positiva

datos_sev <- subset(datos, numclaims > 0 & claimcst0 > 0)

# Coste medio por siniestro

datos_sev$severity <- datos_sev$claimcst0 / datos_sev$numclaims
```

a) Descriptivo de los datos

```
dim(datos)

## [1] 67856      9

kable(summary(datos[, c("veh_value", "exposure",
                        "numclaims", "claimcst0")]))
```

veh_value	exposure	numclaims	claimcst0
Min. : 0.000	Min. :0.002738	Min. :0.00000	Min. : 0.0
1st Qu.: 1.010	1st Qu.:0.219028	1st Qu.:0.00000	1st Qu.: 0.0
Median : 1.500	Median :0.446270	Median :0.00000	Median : 0.0
Mean : 1.777	Mean :0.468652	Mean :0.07276	Mean : 137.3
3rd Qu.: 2.150	3rd Qu.:0.709103	3rd Qu.:0.00000	3rd Qu.: 0.0
Max. :34.560	Max. :0.999316	Max. :4.00000	Max. :55922.1

```
kable(data.frame(table(datos$veh_body)),
      col.names = c("Tipo vehículo", "Frecuencia"))
```

Tipo vehículo	Frecuencia
BUS	48
CONVT	81
COUPE	780
HBACK	18915
HDTOP	1579
MCARA	127
MIBUS	717
PANVN	752
RDSTR	27
SEDAN	22233
STNWG	16261
TRUCK	1750
UTE	4586

```
kable(data.frame(table(datos$veh_age)),
      col.names = c("Edad vehículo", "Frecuencia"))
```

Edad vehículo	Frecuencia
1	12257
2	16587
3	20064
4	18948

```
kable(data.frame(table(datos$gender)),
      col.names = c("Sexo", "Frecuencia"))
```

Sexo	Frecuencia
F	38603
M	29253

```
kable(data.frame(table(datos$area)),
      col.names = c("Área", "Frecuencia"))
```

Área	Frecuencia
A	16312
B	13341
C	20540
D	8173
E	5912
F	3578

```
media_freq <- mean(datos$numclaims)
var_freq <- var(datos$numclaims)
```

```
media_freq
```

```
## [1] 0.07275701
```

```
var_freq
```

```
## [1] 0.07739737
```

La variable numClaims representa el número de siniestros y, por tanto, se trata de una variable de conteo.

b) Modelos de frecuencia

Proponemos también como en la primer parte un modelo de Poisson ($N_i \sim \text{Poisson}(\mu_i)$, donde N_i es el número de siniestros para la póliza i).

```
formula_freq <- numclaims ~ veh_value + veh_body + veh_age +  
  gender + area + agecat + offset(log(exposure))
```

```
modelo_pois <- glm(formula_freq,  
  family = poisson(link = "log"),  
  data = datos)
```

```
summary(modelo_pois)
```

```
##
```

```
## Call:
```

```
## glm(formula = formula_freq, family = poisson(link = "log"), data = datos)
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept) -0.667803   0.326382  -2.046  0.04075 *  
## veh_value    0.023980   0.017251   1.390  0.16451  
## veh_bodyCONVT -1.677911  0.668319  -2.511  0.01205 *  
## veh_bodyCOUPE -0.514222  0.337512  -1.524  0.12762  
## veh_bodyHBACK -0.975986  0.318672  -3.063  0.00219 **  
## veh_bodyHDTOP -0.829563  0.327953  -2.530  0.01142 *  
## veh_bodyMCARA -0.369072  0.409609  -0.901  0.36757  
## veh_bodyMIBUS -0.987654  0.350494  -2.818  0.00483 **  
## veh_bodyPANVN -0.855177  0.339270  -2.521  0.01171 *  
## veh_bodyRDSTR -0.567799  0.660305  -0.860  0.38984  
## veh_bodySEDAN -0.923014  0.318063  -2.902  0.00371 **  
## veh_bodySTNWG -0.906750  0.318239  -2.849  0.00438 **  
## veh_bodyTRUCK -0.944163  0.328476  -2.874  0.00405 **  
## veh_bodyUTE   -1.108652  0.322203  -3.441  0.00058 ***  
## veh_age2      0.054363  0.044623   1.218  0.22312  
## veh_age3     -0.055849  0.048233  -1.158  0.24691  
## veh_age4     -0.114908  0.056801  -2.023  0.04307 *  
## genderM      -0.026181  0.030135  -0.869  0.38495  
## areaB        0.053158  0.042802   1.242  0.21425  
## areaC        0.005108  0.038994   0.131  0.89577  
## areaD       -0.110402  0.052973  -2.084  0.03715 *  
## areaE       -0.031935  0.057878  -0.552  0.58111  
## areaF        0.063729  0.066158   0.963  0.33541  
## agecat2     -0.173250  0.054187  -3.197  0.00139 **  
## agecat3     -0.230051  0.052896  -4.349 1.37e-05 ***  
## agecat4     -0.256934  0.052744  -4.871 1.11e-06 ***  
## agecat5     -0.474475  0.059120  -8.026 1.01e-15 ***  
## agecat6     -0.453279  0.067686  -6.697 2.13e-11 ***
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 25507   on 67855   degrees of freedom
## Residual deviance: 25332   on 67828   degrees of freedom
## AIC: 34823
##
## Number of Fisher Scoring iterations: 6
```

Diagnóstico de sobredispersión:

```
disp_dev <- deviance(modelo_pois) / df.residual(modelo_pois)

disp_pearson <- sum(residuals(modelo_pois,
                             type = "pearson")^2) /
  df.residual(modelo_pois)

disp_dev
```

```
## [1] 0.3734712
```

```
disp_pearson
```

```
## [1] 1.412141
```

La dispersión de Pearson también es relativamente alta en este caso, por lo tanto, probamos nuevamente con una Binomial Negativa:

```
modelo_nb <- glm.nb(formula_freq, data = datos)
summary(modelo_nb)
```

```
##
## Call:
## glm.nb(formula = formula_freq, data = datos, init.theta = 2.279663984,
##      link = log)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  -0.67599    0.34457  -1.962  0.04978 *
## veh_value      0.02514    0.01766   1.424  0.15450
## veh_bodyCONVT -1.67361    0.68249  -2.452  0.01420 *
## veh_bodyCOUPE -0.50686    0.35579  -1.425  0.15427
## veh_bodyHBACK -0.96755    0.33690  -2.872  0.00408 **
## veh_bodyHDTOP -0.82519    0.34616  -2.384  0.01713 *
## veh_bodyMCARA -0.36244    0.42943  -0.844  0.39866
## veh_bodyMIBUS -0.98542    0.36831  -2.676  0.00746 **
## veh_bodyPANVN -0.85238    0.35751  -2.384  0.01712 *
## veh_bodyRDSTR -0.57564    0.68897  -0.835  0.40344
## veh_bodySEDAN -0.91611    0.33630  -2.724  0.00645 **
## veh_bodySTNWG -0.90159    0.33648  -2.679  0.00737 **
## veh_bodyTRUCK -0.94158    0.34663  -2.716  0.00660 **
## veh_bodyUTE   -1.10432    0.34035  -3.245  0.00118 **
## veh_age2       0.05665    0.04564   1.241  0.21445
## veh_age3      -0.05265    0.04929  -1.068  0.28546
## veh_age4      -0.10944    0.05801  -1.887  0.05922 .
```

```
## genderM      -0.02590    0.03078   -0.841   0.40007
## areaB        0.05470    0.04375    1.250   0.21120
## areaC        0.00680    0.03984    0.171   0.86447
## areaD       -0.10811    0.05400   -2.002   0.04531 *
## areaE       -0.02903    0.05906   -0.492   0.62306
## areaF        0.06572    0.06774    0.970   0.33193
## agecat2     -0.17690    0.05553   -3.186   0.00144 **
## agecat3     -0.23271    0.05418   -4.295  1.75e-05 ***
## agecat4     -0.25987    0.05402   -4.811  1.50e-06 ***
## agecat5     -0.47804    0.06042   -7.912  2.53e-15 ***
## agecat6     -0.45779    0.06912   -6.624  3.51e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for Negative Binomial(2.2797) family taken to be 1)
##
##      Null deviance: 23592  on 67855  degrees of freedom
## Residual deviance: 23424  on 67828  degrees of freedom
## AIC: 34786
##
## Number of Fisher Scoring iterations: 1
##
##
##              Theta:  2.280
##              Std. Err.:  0.423
##
## 2 x log-likelihood:  -34727.864
```

Comparación de los modelos:

```
comparacion_freq <- data.frame(
  Modelo = c("Poisson", "Binomial Negativa"),
  AIC = c(AIC(modelo_pois), AIC(modelo_nb)),
  Deviance = c(deviance(modelo_pois), deviance(modelo_nb))
)

kable(comparacion_freq,
      digits = 4,
      booktabs = TRUE)
```

Modelo	AIC	Deviance
Poisson	34822.51	25331.81
Binomial Negativa	34785.86	23424.05

```
if (AIC(modelo_nb) < AIC(modelo_pois)) {
  modelo_freq_final <- modelo_nb
  nombre_modelo_freq <- "Binomial Negativa"
} else {
  modelo_freq_final <- modelo_pois
  nombre_modelo_freq <- "Poisson"
}

nombre_modelo_freq
```

```
## [1] "Binomial Negativa"
```

c) Modelos de severidad

Proponemos un modelo Gamma en primera instancia, siguiendo un procedimiento similar a la primera parte:

```
formula_sev <- severity ~ veh_value + veh_body + veh_age +  
  gender + area + agecat
```

```
modelo_gamma <- glm(formula_sev,  
  family = Gamma(link = "log"),  
  data = datos_sev)
```

```
summary(modelo_gamma)
```

```
##  
## Call:  
## glm(formula = formula_sev, family = Gamma(link = "log"), data = datos_sev)  
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)  
## (Intercept)   6.98190    0.60358  11.567 < 2e-16 ***  
## veh_value     0.02926    0.03562   0.821 0.411522  
## veh_bodyCONVT 0.69403    1.17679   0.590 0.555377  
## veh_bodyCOUPE 0.71276    0.62320   1.144 0.252805  
## veh_bodyHBACK 0.59379    0.58861   1.009 0.313125  
## veh_bodyHDTOP 0.48214    0.60504   0.797 0.425562  
## veh_bodyMCARA -0.65816    0.75190  -0.875 0.381440  
## veh_bodyMIBUS 0.79824    0.64413   1.239 0.215315  
## veh_bodyPANVN 0.57234    0.62643   0.914 0.360948  
## veh_bodyRDSTR -0.99752    1.37406  -0.726 0.467899  
## veh_bodySEDAN 0.44719    0.58768   0.761 0.446727  
## veh_bodySTNWG 0.41023    0.58845   0.697 0.485747  
## veh_bodyTRUCK 0.62943    0.60633   1.038 0.299275  
## veh_bodyUTE   0.50108    0.59477   0.842 0.399567  
## veh_age2      0.07907    0.08059   0.981 0.326559  
## veh_age3      0.12305    0.08787   1.400 0.161497  
## veh_age4      0.20756    0.10492   1.978 0.047956 *  
## genderM       0.17311    0.05457   3.172 0.001523 **  
## areaB        -0.01624    0.07771  -0.209 0.834438  
## areaC         0.09087    0.07096   1.281 0.200426  
## areaD        -0.01070    0.09610  -0.111 0.911334  
## areaE         0.17390    0.10553   1.648 0.099463 .  
## areaF         0.40234    0.12148   3.312 0.000934 ***  
## agecat2      -0.17709    0.09799  -1.807 0.070785 .  
## agecat3      -0.28355    0.09542  -2.972 0.002978 **  
## agecat4      -0.25864    0.09546  -2.709 0.006766 **  
## agecat5      -0.36700    0.10667  -3.441 0.000586 ***  
## agecat6      -0.29438    0.12201  -2.413 0.015876 *  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## (Dispersion parameter for Gamma family taken to be 3.065347)  
##  
##      Null deviance: 7338.6  on 4623  degrees of freedom
```

```

## Residual deviance: 7135.4 on 4596 degrees of freedom
## AIC: 78869
##
## Number of Fisher Scoring iterations: 7

y lo comparamos contra una LogNormal

modelo_lognormal <- lm(log(severity) ~ veh_value + veh_body +
                        veh_age + gender + area + agecat,
                        data = datos_sev)

summary(modelo_lognormal)

##
## Call:
## lm(formula = log(severity) ~ veh_value + veh_body + veh_age +
##     gender + area + agecat, data = datos_sev)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.9390 -0.9284 -0.1741  0.8079  4.2029
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  6.586e+00  4.019e-01  16.385 < 2e-16 ***
## veh_value    1.597e-02  2.372e-02   0.673 0.500882
## veh_bodyCONVT 1.660e-01  7.836e-01   0.212 0.832211
## veh_bodyCOUPE 3.437e-01  4.150e-01   0.828 0.407655
## veh_bodyHBACK 1.788e-01  3.920e-01   0.456 0.648220
## veh_bodyHDTOP 3.915e-02  4.029e-01   0.097 0.922596
## veh_bodyMCARA -4.675e-01  5.007e-01  -0.934 0.350535
## veh_bodyMIBUS 4.159e-01  4.289e-01   0.970 0.332282
## veh_bodyPANVN 1.534e-01  4.171e-01   0.368 0.713172
## veh_bodyRDSTR -6.800e-01  9.150e-01  -0.743 0.457407
## veh_bodySEDAN 9.450e-02  3.913e-01   0.241 0.809197
## veh_bodySTNWG 1.006e-01  3.919e-01   0.257 0.797443
## veh_bodyTRUCK 2.147e-01  4.038e-01   0.532 0.594889
## veh_bodyUTE   1.645e-01  3.961e-01   0.415 0.677986
## veh_age2      8.043e-02  5.367e-02   1.499 0.134015
## veh_age3      1.628e-01  5.852e-02   2.782 0.005432 **
## veh_age4      2.328e-01  6.987e-02   3.331 0.000871 ***
## genderM       8.500e-02  3.634e-02   2.339 0.019383 *
## areaB         8.016e-05  5.175e-02   0.002 0.998764
## areaC         3.427e-02  4.725e-02   0.725 0.468302
## areaD         6.441e-02  6.399e-02   1.006 0.314262
## areaE         1.630e-01  7.028e-02   2.320 0.020400 *
## areaF         2.908e-01  8.090e-02   3.594 0.000329 ***
## agecat2      -1.996e-01  6.525e-02  -3.060 0.002230 **
## agecat3      -2.113e-01  6.354e-02  -3.325 0.000892 ***
## agecat4      -2.238e-01  6.357e-02  -3.521 0.000435 ***
## agecat5      -2.615e-01  7.103e-02  -3.682 0.000234 ***
## agecat6      -1.972e-01  8.125e-02  -2.428 0.015235 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```
## Residual standard error: 1.166 on 4596 degrees of freedom
## Multiple R-squared:  0.01649,    Adjusted R-squared:  0.01071
## F-statistic: 2.854 on 27 and 4596 DF,  p-value: 1.213e-06
```

```
AIC_gamma <- AIC(modelo_gamma)
AIC_lognormal <- AIC(modelo_lognormal)
```

```
comparacion_sev <- data.frame(
  Modelo = c("Gamma", "Lognormal"),
  AIC = c(AIC_gamma, AIC_lognormal)
)
```

```
kable(comparacion_sev,
      digits = 4,
      booktabs = TRUE)
```

Modelo	AIC
Gamma	78869.35
Lognormal	14571.67

```
if (AIC_gamma < AIC_lognormal) {
  modelo_sev_final <- modelo_gamma
  nombre_modelo_sev <- "Gamma"
} else {
  modelo_sev_final <- modelo_lognormal
  nombre_modelo_sev <- "Lognormal"
}
```

```
nombre_modelo_sev
```

```
## [1] "Lognormal"
```

d) Cálculo de la prima pura

La prima pura se calcula mediante:

$$E(S_i) = E(N_i) \cdot E(Y_i)$$

Predicción de la frecuencia anual

```
datos_pred <- datos
datos_pred$exposure <- 1
datos$freq_pred <- predict(modelo_freq_final, newdata = datos_pred, type = "response")
```

Predicción de severidad

```
if (nombre_modelo_sev == "Gamma") {
  datos$sev_pred <- predict(modelo_sev_final,
                           newdata = datos,
                           type = "response")
} else if (nombre_modelo_sev == "Lognormal") {
  datos$sev_pred <- exp(predict(modelo_sev_final,
                               newdata = datos))
}
```

Prima pura final:

```
datos$prima_pura <- datos$freq_pred * datos$sev_pred
```

```
summary(datos$prima_pura)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      48.5   114.1   129.7   136.6   145.3   760.6
```

```
resultado_primas <- data.frame(
  id = 1:nrow(datos),
  veh_body = datos$veh_body,
  veh_age = datos$veh_age,
  gender = datos$gender,
  area = datos$area,
  agecat = datos$agecat,
  frecuencia_predicha = datos$freq_pred,
  severidad_predicha = datos$sev_pred,
  prima_pura = datos$prima_pura
)

kable(head(resultado_primas, 20),
  digits = 4,
  booktabs = TRUE,
  caption = "Resumen de primas puras")
```

Table 9: Resumen de primas puras

id	veh_body	veh_age	gender	area	agecat	frecuencia_predicha	severidad_predicha	prima_pura
1	HBACK	3	F	C	2	0.1589	879.1470	139.6735
2	HBACK	2	F	A	4	0.1619	763.3394	123.5736
3	UTE	2	F	E	2	0.1576	940.2225	148.1901
4	STNWG	2	F	D	2	0.1823	810.5148	147.7939
5	HBACK	4	F	C	2	0.1488	937.7787	139.5648
6	HDTOP	3	M	C	4	0.1683	824.9048	138.7955
7	PANVN	3	M	A	4	0.1610	887.7147	142.8916
8	HBACK	2	M	B	6	0.1382	859.5029	118.7981
9	HBACK	4	F	A	3	0.1391	892.8581	124.1892
10	HBACK	4	F	B	4	0.1425	879.8319	125.3521
11	HBACK	2	M	A	2	0.1729	856.1516	148.0354
12	HBACK	3	M	C	4	0.1431	936.6804	134.0018
13	HBACK	2	F	C	4	0.1629	789.5771	128.5962
14	STNWG	1	M	A	5	0.1489	751.6156	111.9115
15	SEDAN	3	M	B	6	0.1311	860.3896	112.7727
16	SEDAN	2	M	C	4	0.1728	807.3126	139.5435
17	SEDAN	3	F	F	4	0.1651	1026.7199	169.5565
18	HBACK	3	M	C	4	0.1414	929.8253	131.4915
19	HBACK	4	F	D	2	0.1312	959.5462	125.8707
20	HBACK	3	F	C	3	0.1496	866.6356	129.6569

```
write.csv2(resultado_primas,
  file = "primas_practica3_primas_pura.csv",
  row.names = FALSE)
```